

SATHWIK KURMA

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SOFTWARE DEVELOPMENT ENGINEER I

PROFESSIONAL SUMMARY

Software Engineer with hands-on experience building distributed, cloud-native backend systems like microservices, event-driven architecture, and REST APIs across ASP.NET Core and Python. Strong foundation in data structures & algorithms (400+ LeetCode problems solved, rating 1643) and a track record of designing systems for fault tolerance, concurrency, and horizontal scale. Top 3% nationally at Amazon ML Summer School.

TECHNICAL SKILLS

Languages: Python, C#, C++, TypeScript, JavaScript, SQL

Backend & Data: FastAPI, ASP.NET Core, REST APIs, Entity Framework Core, PostgreSQL, SQL Server, Redis

Distributed Systems: Microservices, RabbitMQ, Message Queues, Event-Driven Architecture, API Gateway (YARP)

Cloud/DevOps: Docker, Docker Compose, AWS, Git, CI basics, Airflow

Core CS: Data Structures & Algorithms, OOP, System Design Fundamentals, Design Patterns, Agile/SDLC

EXPERIENCE

Capgemini

Dec 2025 - May 2026

Trainee Engineer

- Build backend services in ASP.NET Core and SQL Server following clean-architecture principles, delivering [15] REST API endpoints consumed by an Angular front end across [6] internal modules.
- Implement authentication workflows (JWT-based) and containerize services with Docker, reducing environment setup time for new developers by [65]%.

Amazon ML Summer School

Aug 2025 - Sept 2025

Machine Learning Trainee · Top 3% Nationally

Remote

- Built and evaluated ML models (XGBoost, SVM, Random Forest) achieving up to 95% accuracy on classification tasks, as part of a nationally competitive cohort.
- Completed advanced coursework in statistical machine learning and large-scale system design, ranking in the top 3% of participants nationwide.

Infosys Springboard

Oct 2024 - Jan 2025

Project Intern

Remote

- Developed NLP pipelines in Python using SQL and FAISS for semantic retrieval over financial documents.
- Improved semantic search precision by 35% by tuning embedding and indexing strategy for the retrieval pipeline.

PROJECTS

Distributed Task Scheduler | Fault-Tolerant Job Queue

July 2026

Python, FastAPI, Redis, PostgreSQL, Docker Compose

GitHub

- Built a distributed job queue with a FastAPI producer and horizontally scalable worker pool, using blocking *BLPOP* for zero-latency, atomic job dequeuing across concurrent workers.
- Engineered at-least-once delivery via a TTL-based processing lock plus a reaper thread that auto-recovers jobs from crashed workers within ~90 seconds without manual intervention.
- Implemented idempotent job submission with a database-enforced unique constraint, preventing duplicate execution under concurrent retries; added exponential-backoff retries with a dead-letter queue for exhausted failures.
- Load-tested the distributed scheduler with 50 concurrent users using Locust, achieving **82 requests/sec** while maintaining a **1.3 s p99 latency**; covered core logic with pytest, fakeredis, and SQLite integration tests.

Mystic Tarot | Full-Stack Digital Consultation Platform

Ongoing

ASP.NET Core 10 | Angular 17 | SQL Server | Docker | RabbitMQ | OpenAI API

GitHub

- Engineered a 5-service ASP.NET Core microservices backend with a YARP API Gateway, JWT role-based authentication, and EF Core with a database-per-service architecture.
- Integrated OpenAI GPT-4o-mini for AI-powered tarot readings and built a RabbitMQ-based event-driven notification system with graceful failure handling.
- Built an Angular 17 SPA with standalone components, Signals, and route guards; containerized 8 services with Docker Compose and automated database migrations.

CraveKart | Microservices Food Delivery Platform

April 2026

ASP.NET Core | RabbitMQ | Docker | SQL Server | API Gateway | C# | Angular 17

GitHub

- Designed a distributed microservices backend with RabbitMQ-based async communication and API Gateway routing for centralized auth and request orchestration; containerized all services with Docker Compose.

EDUCATION

Lovely Professional University

2022 - 2026

B.Tech in Computer Science & Engineering · CGPA: 8.01/10

Jalandhar, Punjab

CERTIFICATIONS & ACHIEVEMENTS

- Amazon ML Summer School — Top 3% Nationally (Aug-Sept 2025)
- Solved 400+ LeetCode problems (Rating: 1643); Advanced SQL 4-star, 100% score on HackerRank
- Supervised Machine Learning: Regression & Classification - DeepLearning.AI